

An Exact Chern-Phase Atlas for the QWZ Thouless Pump

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Abstract

For a two-band pump on the Bloch torus we prove the exact spectrum, direct gap, all lower-band Chern phases, and every gap-closing Dirac jump. The transported charge statement is made only in the filled-band adiabatic limit. Independent symbolic, degree, and lattice-gauge receipts audit the convention-sensitive signs. This revision is the **exact Bloch-gap owner**.

1 Model and complete gap atlas

Orient $\mathbb{T}^2$ by $dk \wedge d\tau$ and define

$$H_m(k, \tau) = d_m(k, \tau) \cdot \sigma, \quad d_m = (\sin k, \sin \tau, m + \cos k + \cos \tau). \quad (1)$$

Here $m \in \mathbb{R}$ and both angular variables have period 2π . Whenever d_m is nonzero, let $n = d_m/|d_m|$ and $P_- = (I - n \cdot \sigma)/2$.

Theorem 1 (Spectrum, gap, and lower-band phase). *The fibre spectrum is $\{-|d_m|, +|d_m|\}$. The family is gapped exactly for $m \notin \{-2, 0, 2\}$, with direct gap*

$$G(m) = 2 \min\{|m+2|, |m|, |m-2|\}.$$

On a gapped chamber P_- is a smooth rank-one complex line bundle. With

$$c_1(P_-) = \frac{1}{2\pi i} \int_{\mathbb{T}^2} \text{Tr}(P_- [\partial_k P_-, \partial_\tau P_-]) dk d\tau, \quad (2)$$

its Chern number is

$$c_1 = \begin{cases} 0, & m < -2, \\ -1, & -2 < m < 0, \\ +1, & 0 < m < 2, \\ 0, & m > 2. \end{cases}$$

Proof of the spectral and gap assertions. Pauli multiplication gives $H_m^2 = |d_m|^2 I$. With $x = \cos k$ and $y = \cos \tau$,

$$|d_m|^2 = m^2 + 2 + 2m(x + y) + 2xy. \quad (3)$$

The right side is affine in each variable separately, so its minimum on $[-1, 1]^2$ occurs at a corner. The four corner values are $(m+2)^2, m^2, m^2, (m-2)^2$. This proves both the closing locus and the stated gap. Smooth functional calculus then gives the displayed lower projector. The Chern computation is proved below in revision one. $\square$

2 Projector sign, degree, and Dirac walls

This revision is the **analytic Dirac-degree owner**. Direct Pauli algebra fixes the convention-sensitive sign:

$$\text{Tr}(P_- [\partial_k P_-, \partial_\tau P_-]) = -\frac{i}{2} n \cdot (\partial_k n \times \partial_\tau n). \quad (4)$$

Consequently (2) is minus the Brouwer degree of $n : \mathbb{T}^2 \rightarrow S^2$. For completeness, a direct differentiation before normalization gives

$$d_m \cdot (\partial_k d_m \times \partial_\tau d_m) = \cos k + \cos \tau + m \cos k \cos \tau. \quad (5)$$

Choose the north pole as regular value away from the walls. Its possible preimages are the four points with $\sin k = \sin \tau = 0$. A point contributes exactly when its mass $M = m + \cos k_0 + \cos \tau_0$ is positive, and its local degree is $\chi(k_0, \tau_0) = \cos k_0 \cos \tau_0$. Thus $\deg(n) = \sum \chi \mathbf{1}_{\{M>0\}}$. Substituting $\mathbf{1}_{\{M>0\}} = (1 + \operatorname{sgn} M)/2$ and using $\sum \chi = 0$ yields

$$c_1 = -\frac{1}{2} [\operatorname{sgn}(m+2) - 2\operatorname{sgn}(m) + \operatorname{sgn}(m-2)]. \quad (6)$$

This proves the remaining part of Theorem 1 without numerical integration. As m increases, the complete wall ledger is

mass	zero (k_0, τ_0)	χ	jump of c_1
-2	(0, 0)	+1	-1
0	(π , 0)	-1	+1
0	(0, π)	-1	+1
2	(π , π)	+1	-1

The simultaneous pair at $m = 0$ is essential: its total jump is two.

3 Adiabatic transport and exact boundaries

This revision is the **adiabatic scope and route firewall**.

Corollary 1 (Filled-band Thouless transport). *Let τ increase from 0 to 2π , assign unit positive particle charge, and define positive current by $J = \partial_k H_m$. Assume a gapped chamber, a completely filled lower band, and an adiabatic traversal. With the orientation fixed in (2), the transported positive-particle charge per cycle equals $c_1(P_-)$.*

Indeed, for

$$\Omega_{k\tau} = i \operatorname{Tr} P_- [\partial_k P_-, \partial_\tau P_-], \quad Q = -\frac{1}{2\pi} \int_{\mathbb{T}^2} \Omega_{k\tau} dk d\tau = \frac{1}{2\pi i} \int_{\mathbb{T}^2} \operatorname{Tr} P_- [\partial_k P_-, \partial_\tau P_-] = c_1(P_-).$$

For electronic charge the result is multiplied by $-e$. This is an adiabatic-limit statement. At finite driving rate, interband transitions can occur; no exact finite-driving-rate quantization is claimed.

At $m = -2, 0, 2$ the lower projector is undefined at the zeroes in the table and no gapped-bundle Chern number is assigned. Reversing the torus orientation, or sending $\tau \mapsto -\tau$, reverses the integer. Complex conjugation satisfies

$$KH_m(k, \tau)K = H_m(k, -\tau),$$

because only σ_y is imaginary. For $m > 2$ (respectively $m < -2$), the z component is everywhere positive (respectively negative), and the field contracts without closing the gap to the north (respectively south) pole.

4 Evidence, provenance, and claim boundary

The exact evidence records the four corner values, chamber signs, and Dirac ledger. A separate discrete lattice-gauge lane on several grids reproduces the four integers, but is only regression evidence; the analytic degree proof owns the theorem. Strict parsers, repaired-hash hostile mutations, isolated byte replay, and deterministic PDF builds audit the release.

Thouless established quantized adiabatic particle transport [1]; Qi, Wu, and Zhang give the two-band lattice-Dirac lineage [2]. We cite lineage, not priority, and rederive every normalization and sign used here. Nearby packages treat finite SSH chains, the Dirac-monopole sphere and its magnetic spectrum, and kicked rotors; none owns this smooth two-parameter Bloch-bundle atlas.

The Chern integer is a source-local natural quantization, but the Chern integer is not a rational-prime carrier. There is no primitive-orbit determinant, target functional equation, target zero match, or Hilbert-Pólya operator. Thus Route A is rejected as (`A0_FAIL`, `A1_FAIL`, `A2_FAIL`, `A3_FAIL`, `A4_NATURAL_QUANTIZATION`), and Route B remains locked under `NO_BAD_EULER_OR_ROOT_NUMBER`.

References

- [1] D. J. Thouless, *Quantization of particle transport*, Phys. Rev. B 27 (1983), 6083–6087. <https://doi.org/10.1103/PhysRevB.27.6083>.
- [2] X.-L. Qi, Y.-S. Wu, and S.-C. Zhang, *Topological quantization of the spin Hall effect in two-dimensional paramagnetic semiconductors*, Phys. Rev. B 74 (2006), 085308. <https://doi.org/10.1103/PhysRevB.74.085308>.